Launch One ec2 instance and connect with mobaXterm → sethostname Terraform

Got to Github devopsScript by reayaz → terrafomAL2.sh

Copy the code and create the fle in the ec2-instance Paste it→ name as terr.sh

Run → sh terr.sh

rm -rf *

clear

Ls -al

```sh
terraform init
```

```sh
cat <<EOF > main.tf
provider "aws" {
  region = "ap-south-1"
}
EOF
```

```sh
terraform init
```
```sh
cat <<EOF > main.tf
variable "fruits" {
  type    = list(string)
  default = ["apple", "banana", "mango"]
}

output "list_length" {
  value = length(var.fruits)
}
EOF
```

```sh
terraform validate
```

```sh
terraform apply --auto-approve
```

```

```sh
cat <<EOF > main.tf
variable "fruits" {
  type    = list(string)
  default = ["apple", "grapes"]
}

variable "vegies" {
  type    = list(string)
  default = ["onions", "tomatoes"]
}

output "combined_list" {
  value = concat(var.fruits, var.vegies)
}
EOF
```

terraform apply --auto-approve

```
cat <<EOF > main.tf
variable "fruits" {
  type    = list(string)
  default = ["apple", "cherry", "kiwi"]
}

output "selected_element" {
  value = element(var.fruits, 1)
}
EOF
```sh
cat <<EOF > main.tf
variable "keys" {
  type    = list(string)
  default = ["name", "age"]
}

variable "values" {
  type    = list(any)
  default = ["puneeth", 22]
}

output "my_map" {
  value = zipmap(var.keys, var.values)
}
EOF
```

```
```sh
cat <<EOF > main.tf
variable "keys" {
  type    = list(string)
  default = ["name", "age"]
}

variable "values" {
  type    = list(any)
  default = ["puneeth", 22]
}

output "my_map" {
  value = zipmap(var.keys, var.values)
}
EOF
```
```sh
cat <<EOF > main.tf
variable "my_map" {
  type    = map(string)
  default = {
    "name" = "venky"
    "age"  = "50"
  }
}

output "value" {
  value = lookup(var.my_map, "name")
}
EOF
```
```sh
cat <<EOF > main.tf
variable "fruits" {
  type    = list(string)
  default = ["apple", "mangoes", "orange"]
}

output "joined_string" {
  value = join(",", var.fruits)
}
EOF
```
cat <<EOF > main.tf
variable "unique_names" {
  type   = set(string)
```

```
  default = ["Alice", "Bob", "Charlie"]
}

output "unique_names_list" {
  value = tolist(var.unique_names)
}
EOF
```sh
cat <<EOF > main.tf
variable "fruits" {
  type    = list(string)
  default = ["apple", "mangoes", "orange"]
}

output "joined_string" {
  value = join(",", var.fruits)
}
EOF
```
```sh
cat <<EOF > main.tf
variable "instance_tags" {
  type = map(string)
  default = {
    Name = "WebServer"
    Env  = "Production"
  }
}

output "instance_tags_output" {
  value = var.instance_tags
}

output "instance_name" {
  value = var.instance_tags["Name"]
}

output "tags_keys" {
  value = keys(var.instance_tags)
}

output "tags_values" {
  value = values(var.instance_tags)
}
EOF
```
```

Vi main.tf
```
```hcl
provider "aws" {
  region = "ap-south-1"
}

variable "instance_count" {
  description = "number of ec2 instances"
  type        = number
  default     = 3
}

variable "instance_ami" {
  description = "AMI to be used"
  type        = string
  default     = "ami-08fe5144e4659a3b3"
}

variable "instance_type" {
  description = "Instance type to be used"
  type        = string
  default     = "t2.micro"
}

variable "instance_name" {
  description = "Name of the instance"
  type        = string
  default     = "TF-Server"
}

resource "aws_instance" "myinstance" {
  count         = var.instance_count
  ami           = var.instance_ami
  instance_type = var.instance_type

  tags = {
    Name = var.instance_name
  }
}
```

```

Vi variables.tf

```hcl
variable "instance_count" {
  description = "number of ec2 instances"
  type        = number
  default     = 3
}

variable "instance_ami" {
  description = "AMI to be used"
  type        = string
  default     = "ami-08fe5144e4659a3b3"
}

variable "instance_type" {
  description = "Instance type to be used"
  type        = string
  default     = "t2.micro"
}

variable "instance_name" {
  description = "Name of the instance"
  type        = string
  default     = "TF-Server"
}
```

Delete all variables from main.tf file
Only keep provider resources section in main.tf file
Then terraform validate , terraform plan , terraform apply –auto-approve

Remove all the values of variable from variable.tf Then create vi teraform.tfvars file add all value in new file called terraform.tfvars

Add all variables like →

```hcl
instance_count = 1
instance_ami   = "ami-08fe5144e4659a3b3"
instance_type  = "t2.micro"
instance_name  = "TF-Server"
```